

D-port static / driven — run report

RUN

191733_first-npbs-d-test_first-npbs-d-test

Status: completed | Started: 2026-09-18T18:17:33.713116+00:00

PURPOSE

Compare static polarization with a phi1 sine at first NPBS D.

REQUIRED OPTICAL SETUP

PAX at first-NPBS D port (requires moving it from the current C position). Photodiode at final F on the configured RP input.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port

Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

DATA

715 recorded samples; static: 359, phi1_sine: 356

Source: data.csv; full acquisition settings: recipe.json; status: run.json

RAW DOP

Mean 0.8668; range 0.5331 to 0.9221. 0 samples outside 0-1. Telemetry is not a new calibration.

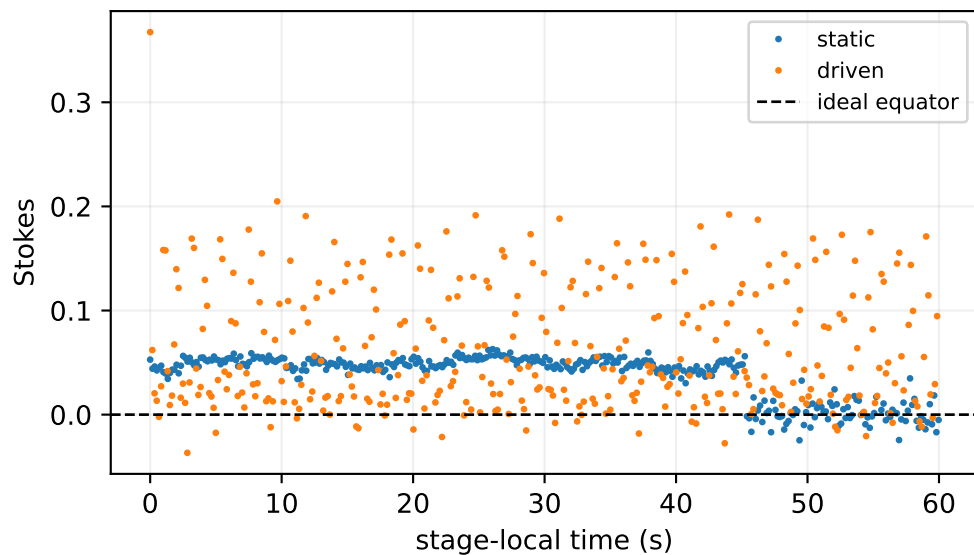
PLACEMENT NOTE

The recipe records C, but this test's instructions specify D. Actual placement was not independently logged; confirm it before interpreting the D-port model.

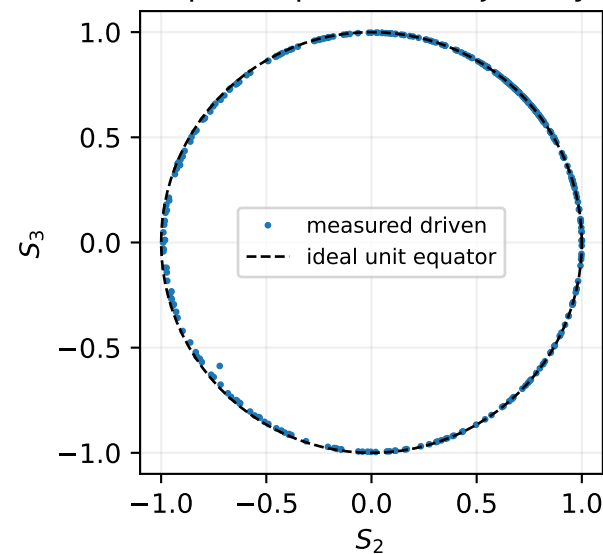
First-NPBS D-port: polarization response

Drive: 0.50 Hz | Equator S1 RMS: 0.0865 | Nominal trajectory RMS: 1.3206

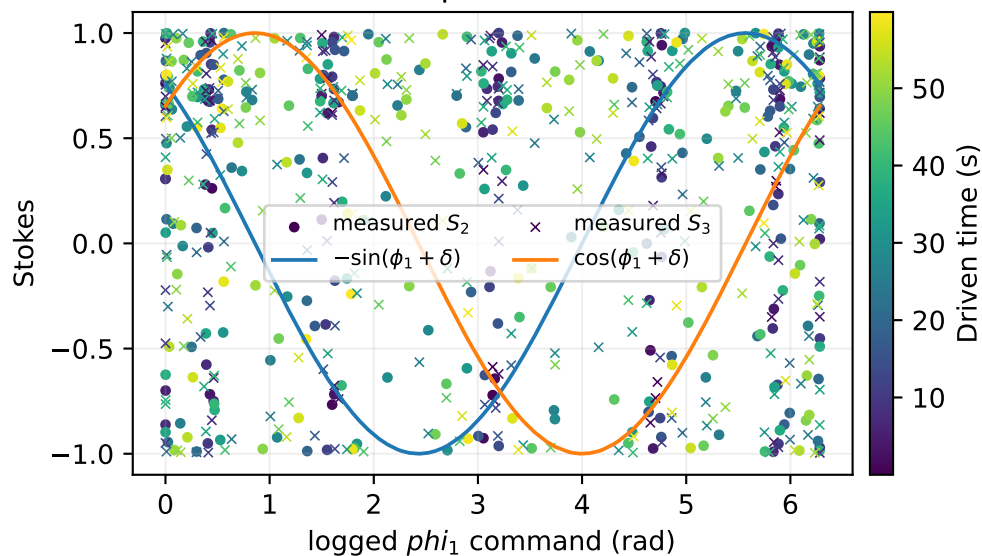
Equator test: ideal $S_1 = 0$



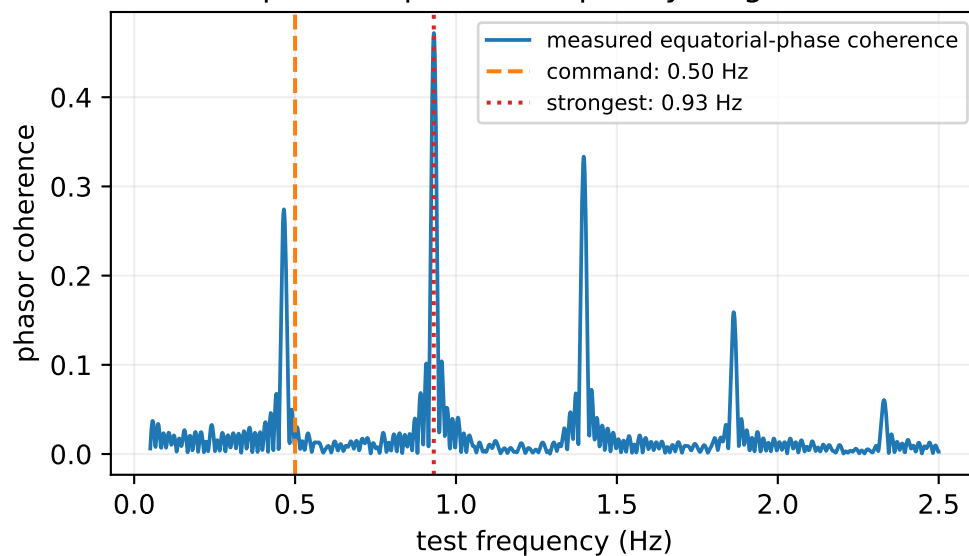
D-port equatorial trajectory



Command-to-polarization transfer



Equatorial phasor frequency diagnostic

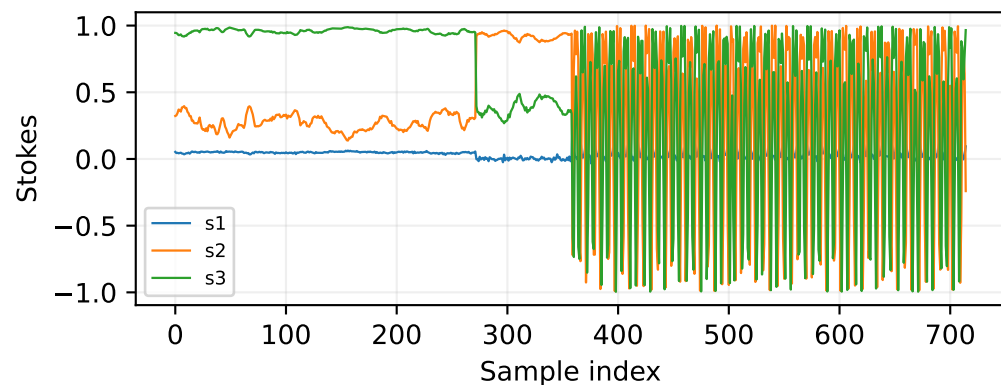


Command phase uses the configured V_{λ} and estimated waveform timing; this is not a measured V_{π} fit.

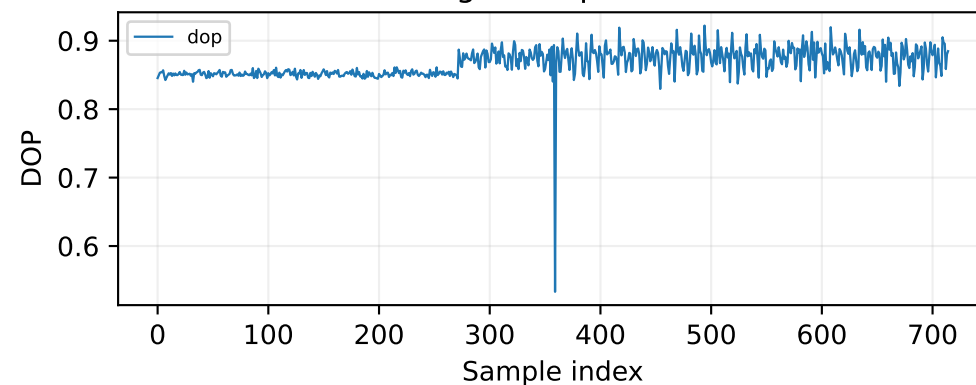
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

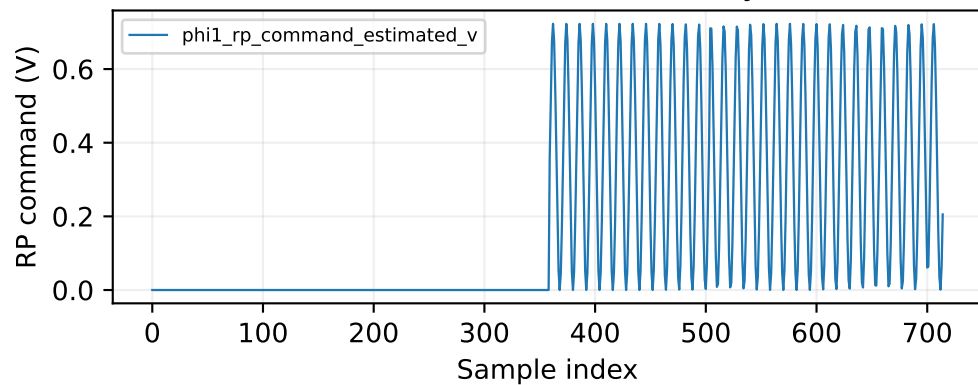
Normalized Stokes



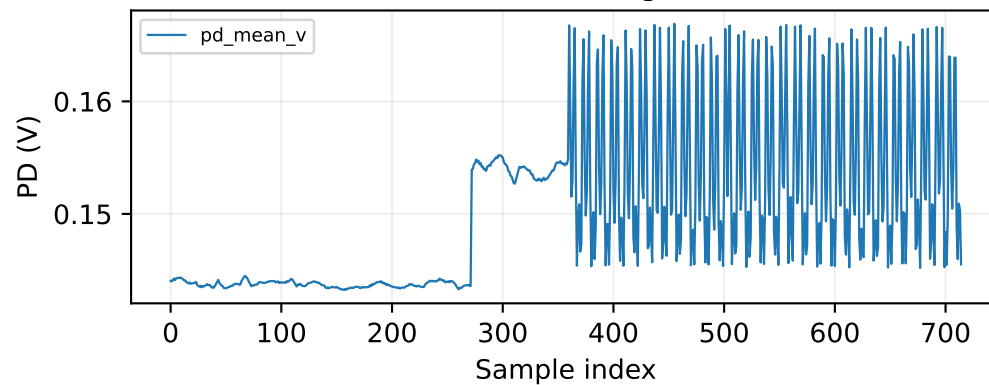
Raw degree of polarization



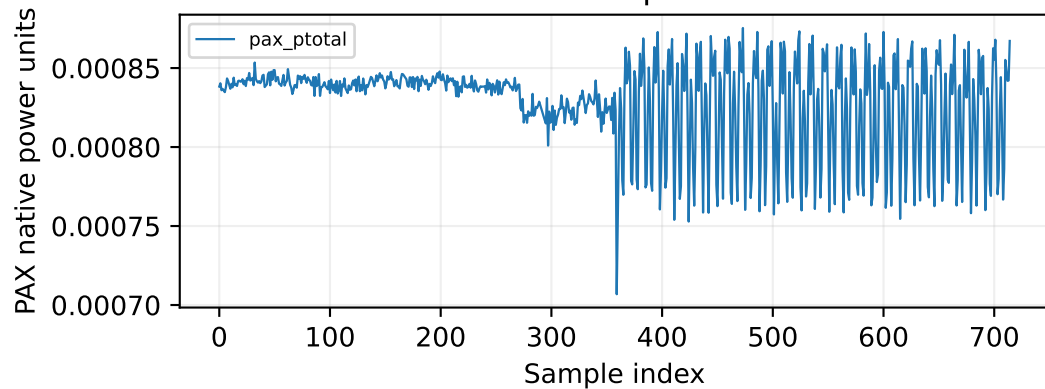
Actuator command history



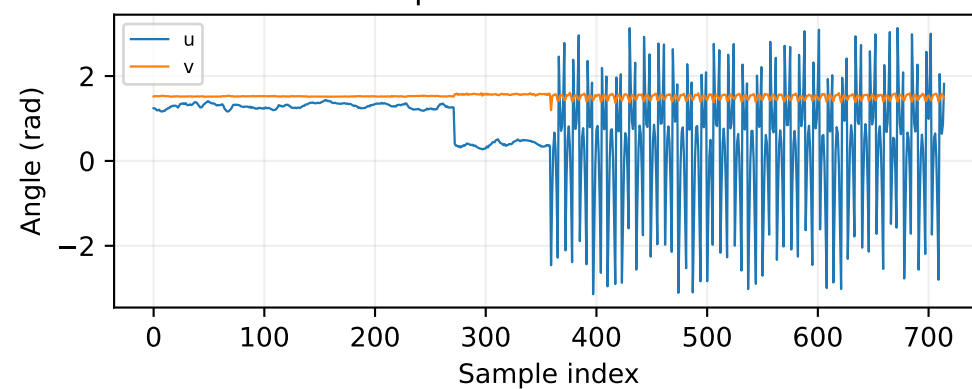
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"first-npbs-d-test"
report	"pdf"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	12.2
phi2_v_lambda	30.0
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	150.0
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0
pd_input	"in1"
pd_nd_optical_density	2.0
pd_scope_duration_s	0.01
pd_scope_decimation	64
pd_scope_timeout_s	2.0
first_npbs_d_static_duration_s	60.0
first_npbs_d_driven_duration_s	60.0
first_npbs_d_phi1_frequency_hz	0.5
first_npbs_d_sample_period_s	0.05